

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Mixed

01

10 answers · Rationale-rich · Teach from doc

Q1

D

- D. the family's division into northern and southern branches likely preceded the acquisition of the crop by the Uto-Aztec tribes.

Choice D is the best answer because it most logically completes the discussion of Uto-Aztec languages. The text explains that the northern and southern branches of the Uto-Aztec language family descended from a single language (believed to have originated in what is now the US Southwest), resulting in similarities across the family's languages; however, the branches don't have similar vocabulary for maize, even though maize has been cultivated by all Uto-Aztec tribes. The text also indicates that maize originated in Mexico and spread northward into what is now the US Southwest—the area where the Uto-Aztec language family originated. It follows, then, that the language family had already divided into northern and southern branches before maize reached that area; if maize had been present before the division occurred, the family's origin language would have had terminology for it that likely would have been reflected in the branches, meaning they would have had similar vocabulary for maize. If maize arrived after the division occurred, however, the tribes in the two regions likely would have developed vocabulary pertaining to maize separately, at the times when they acquired the crop.

Choice A is incorrect because the text focuses on vocabulary pertaining to maize in the branches of the Uto-Aztec language family, and referring only to how some Uto-Aztec tribes obtained maize wouldn't directly address the role of language. Moreover, if northern Uto-Aztec tribes had acquired maize from a southern Uto-Aztec tribe, it's reasonable to assume that the northern tribes might have also picked up southern Uto-Aztec terminology for maize in that exchange. Choice B is incorrect because the text discusses the fact that the northern and southern branches of the Uto-Aztec language family don't have shared vocabulary pertaining to maize, not the idea that there are variations in such vocabulary within each branch—that is, the text focuses on differences between the two branches, not on differences between languages within a branch. Choice C is incorrect because the text focuses on vocabulary pertaining to maize in the branches of the Uto-Aztec language family, and referring only to the timing and source of maize acquisition wouldn't directly address the role of language. Furthermore, the text implies that southern Uto-Aztec tribes probably acquired maize before the northern tribes did, given the evidence that maize originated in Mexico—the location of the best-known representative of the southern branch of the Uto-Aztec language family—before spreading to the north.

Q2

C

- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
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Choice C is the best answer. The text says that the cerebellum has long been thought to regulate motor functioning, but new research shows that it may also have other functions—including regulating emotion and social behavior and some connection to motivation and rewards processing.

Choice A is incorrect. The VTA is described as “a center for motivation and reward processing,” and the discovery of the pathway between the VTA and the cerebellum supports the theory that the cerebellum is involved in functions other than motor coordination. Choice B is incorrect. The text says that recent research has identified this pathway, but it doesn’t discuss any previous “widely accepted claims” about the pathway’s function. The “widespread acknowledgement” mentioned early in the passage is about the cerebellum alone, not its connection to the VTA. Choice D is incorrect. The text never discusses any technological limitations or any new technologies.

Q3

A

- A. *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.
-

Choice A is the best answer because it most logically completes the text's discussion of the *M. robustus* population at the Octopus Garden. The text states that the scientists concluded that the site is likely used only for reproduction because over three years they saw many adults, freshly hatched octopuses, and eggs but didn't see any juveniles. This suggests that the *M. robustus* octopuses that hatch at the Octopus Garden leave the site when they reach an intermediary state of development, returning only as adults for reproductive purposes.

Choice B is incorrect because the text never discusses the stability of the *M. robustus* population at the site, only that the scientists observed 6,000 adults, hatchlings, and eggs there. Further, the text presents the site's temperatures as likely beneficial. Choice C is incorrect because the text doesn't provide any details about the eggs at the site and makes no mention of nests; it indicates only that eggs are present along with hatchlings and adults. Choice D is incorrect because the text makes no mention of the hatchlings feeding at the Octopus Garden, indicating only that the temperatures at the site are probably beneficial and that the site is likely used for reproduction.

Q4

D

- D. “Alexandra drew her shawl closer about her and stood leaning against the frame of the mill, looking at the stars which glittered so keenly through the frosty autumn air. She always loved to watch them, to think of their vastness and distance, and of their ordered march. It fortified her to reflect upon the great operations of nature, and when she thought of the law that lay behind them, she felt a sense of personal security.”
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Choice D is the best answer because it most effectively uses a quotation from *O Pioneers!* to illustrate the claim that Alexandra Bergson takes comfort in understanding the world around her. In the quotation, Alexandra is described as enjoying looking at the stars and feeling a “sense of personal security” when she contemplates nature’s order and its governing laws. This suggests that Alexandra takes comfort in understanding the world around her.

Choice A is incorrect because the quotation expresses how Alexandra Bergson attempts to meet difficult situations with determination, not how she takes comfort in understanding the world around her. Choice B is incorrect because the quotation expresses “how much the country meant to” Alexandra Bergson, not how she takes comfort in understanding the world around her. In detailing some of the wildlife surrounding Alexandra, the quotation conveys that nature is important to her but not necessarily that it gives her comfort. Choice C is incorrect because the quotation describes Alexandra driving her wagon down a highway at night; it doesn’t describe how she takes comfort in understanding the world around her or address how she’s feeling as she drives off.

Q5

C

- C. Locke-Rodriguez and colleagues found that FTWs using marigolds not only helped remove phosphorus from the water but also yielded market-quality blooms.
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Choice C is the best answer. The sentence presents both findings from the study, noting that FTWs using marigolds both removed phosphorus from the water and yielded market-quality blooms.

Choice A is incorrect. While the sentence presents a finding from the study, the presentation is incomplete; the study found that marigolds cultivated on FTWs both removed phosphorus from the water and yielded market-quality blooms. Choice B is incorrect. The sentence misrepresents information in the notes. The figure of 52% refers to the amount of phosphorus removed by FTWs using marigolds when compared to the control; the marigolds' bloom yield was 65 market-quality blooms per square meter, and it was not compared to the control. Choice D is incorrect. The sentence misrepresents information in the notes. The FTWs using marigolds removed 52% more phosphorus from the water than did the control, not 52% of the total phosphorus. Additionally, the bloom figure is incomplete; the figure is 65 market-quality flower blooms per square meter.

Q6

B

B. Hence,

Choice B is the best answer. "Hence" correctly signals that the claim in this sentence regarding extraterrestrial material follows logically from the information in the previous sentences. The previous sentences establish that siderophile elements in the mantle are more abundant than predicted in the core-differentiation model. This sentence claims, logically, that these elements came from extraterrestrial material that accreted to Earth *after* core differentiation.

Choice A is incorrect because "that said" illogically signals that the information in this sentence regarding extraterrestrial material is an exception to the previous information about siderophiles' abundance in the mantle. Instead, it is a new claim that follows logically from the previous information. Choice C is incorrect because "for example" illogically signals that the information in this sentence regarding extraterrestrial material exemplifies the previous information about siderophiles' abundance in the mantle. Instead, it is a new claim that follows logically from the previous information. Choice D is incorrect because "likewise" illogically signals that the information in this sentence regarding extraterrestrial material is merely similar to the previous information about siderophiles' abundance in the mantle. Instead, it is a new claim that follows logically from the previous information.

Q7

C

C. audience. To

Choice C is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period is used correctly to mark the boundary between one sentence ("In modern...audience") and another ("To do...performance").

Choice A is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "as." Choice B is incorrect because it results in a run-on sentence. The sentences ("In modern...audience" and "To do...performance") are fused without punctuation and/or a conjunction. Choice D is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences.

Q8

B

B. and, in the process,

Choice B is the best answer. This choice uses paired punctuation in the form of two commas to set off the nonessential phrase "in the process."

Choice A is incorrect. The phrase "in the process" is a nonessential element and needs to be set off with paired punctuation. We need a comma after "and" to match the one after "process."

Choice C is incorrect. The phrase "in the process" is a nonessential element and needs to be set off with paired punctuation, so we would need a dash after "and" to match the one following "process." Choice D is incorrect. The phrase "in the process" is a nonessential element and needs to be set off with paired punctuation. We would need a comma after "process" to match the one following "and."

Q9

B

B. superseded

Choice B is the best answer because it most logically completes the text's discussion of the fork fern's enormous genome. In this context, "superseded" means surpassed or exceeded by something else. The text establishes that this plant has "the largest known" genome among eukaryotes and mentions "the high biological costs of extremely large genomes," suggesting there are natural constraints that make it biologically disadvantageous to maintain such massive genetic material. This context supports the idea that it's unlikely another organism will evolve an even larger genome that would replace this plant's record—that is, its status as having the largest genome is unlikely to be superseded.

Choice A is incorrect because in this context, "protracted" would mean drawn out over time or space, and the text focuses on the record size of the New Caledonian fork fern's genome and the fact that extremely large genomes carry "high biological costs." This information suggests that the biological upper limit for genome size may have been reached and that the New Caledonian fork fern's genome is therefore unlikely to be surpassed, or superseded, not that it's unlikely to be protracted. Choice C is incorrect because in this context, "curtailed" would mean cut short or reduced, and the text focuses on the record size of the New Caledonian fork fern's genome and the fact that extremely large genomes carry "high biological costs." This information suggests that the biological upper limit for genome size may have been reached and that the New Caledonian fork fern's genome is therefore unlikely to be surpassed, or superseded, not that it's unlikely to be curtailed. Choice D is incorrect because in this context, "obviated" would mean made unnecessary or redundant, and the text focuses on the record size of the New Caledonian fork fern's genome and the fact that extremely large genomes carry "high biological costs." This information suggests that the biological upper limit for genome size may have been reached and that the New Caledonian fork fern's genome is therefore unlikely to be surpassed, or superseded, not that it's unlikely to be obviated.

Q10

A

A. sanguine

Choice A is the best answer because it most logically completes the text's discussion of political blogs. In this context, "sanguine" means optimistic. The text begins by noting the rise of political blogs with readily identifiable ideological alignments in the early 2000s. The text then indicates that some commentators saw this as a positive development, citing a reason why (their difference from traditional news). Finally, the text goes on to contrast those commentators with others who have a negative opinion of the rise of political blogs (because they increase political polarization among their readers). This context supports the idea that the second group of commentators is less positive than the first: thus, the second group of commentators is less optimistic, or sanguine.

Choice B is incorrect because it would not make sense in this context to describe those commentators who have a negative opinion of political blogs as less "recalcitrant," or obstinately uncooperative, than those commentators who supported political blogs. Choice C is incorrect because the text gives no indication that those commentators who have a negative opinion of political blogs are less "misanthropic," or less contemptuous of humankind, than those commentators who have a positive opinion of political blogs—there is no indication in the text that those commentators who like political blogs would be contemptuous of humankind at all. Choice D is incorrect because there is no evidence that those commentators who have a negative opinion of political blogs are less "earnest," or sincere, than those who have a positive opinion of such blogs—presumably, both groups of commentators hold their beliefs with equal conviction.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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